National Driving School & Testing Portal (NDSTP)

A Centralized Web Application for Driver Examination Booking and Driving School Directory in Botswana

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Submission Date: October 26, 2026

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Declaration

We, the undersigned, hereby declare that this report entitled *"National Driving School & Testing Portal (NDSTP)"* is our own original work and has not been submitted elsewhere for any academic award or qualification. All sources used have been duly acknowledged and referenced according to the Harvard referencing style. This work has been carried out under the supervision of Mr. Resego Rabolane.

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Acknowledgements

We would like to express our sincere gratitude to our supervisor, Mr. Resego Rabolane, for his invaluable guidance, constructive feedback, and continuous support throughout this project. We also extend our appreciation to the Department of Road Transport and Safety (DRTS) staff and the driving school managers who participated in our interviews and questionnaires. Finally, we thank our families and colleagues for their encouragement and support.

Abstract

The National Driving School & Testing Portal (NDSTP) project addresses the inefficiencies of the current manual driver licensing system in Botswana, which is characterized by long physical queues, limited visibility of available test dates, and a fragmented driving school market. This report presents the design, development, and evaluation of a centralized, secure web-based application that digitizes driver license examination bookings (theory, yard, and road tests) while providing a national directory for accredited driving schools. The system was developed using an Agile, sprint-based methodology over a 14-week timeline, incorporating Role-Based Access Control (RBAC), a real-time slot booking engine with atomic transaction handling, and a two-sided marketplace for driving schools. The Technology Acceptance Model (TAM) was used as the theoretical framework to evaluate user adoption. The prototype was tested through unit and integration testing and User Acceptance Testing (UAT). Results indicate that the portal significantly reduces booking friction, prevents schedule collisions, and improves information symmetry. The report concludes with recommendations for future enhancements.

Keywords: E-Government, Driver Licensing, Automated Scheduling, Two-Sided Platforms, Role-Based Access Control, Technology Acceptance Model.

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Chapter 1

Introduction

1.1 Background of the Study

The National Driving School & Testing Portal (NDSTP) manages driver licensing, which requires students to complete three sequential tests: mainly theory, yard, and road (Government of Botswana, 2020). Currently, booking these test slots involves manual processes, long physical queues, and limited visibility of available dates (World Bank, 2021). At the same time, prospective drivers struggle to find reliable driving schools due to a lack of a central information platform (Mpofu and Ohei, 2022). The NDSTP Web Application solves both issues by providing a centralized system for booking test slots while allowing accredited driving schools to advertise their services directly to learners.

1.2 Problem Statement

The current driver licensing workflow in Botswana is predominantly manual, decentralized, and prone to inefficiencies. Candidates must physically visit DRTS offices to book test slots, often waiting in long queues only to find that desired dates are unavailable. There is no real-time visibility of examiner schedules or slot availability, leading to repeated visits and wasted time. Simultaneously, the driving school sector operates independently, with no centralized platform for learners to verify accreditation, compare fees, or locate schools. This information asymmetry disadvantages both learners and smaller driving schools.

1.3 Research Aim and Objectives

1.3.1 Research Aim

The primary aim of the NDSTP project is to design and build a secure, centralized web application that digitizes driver license examination bookings and provides a national directory for driving schools in Botswana.

1.3.2 Research Objectives

- To analyze current DRTS test allocation workflows and compile feature requirements for both candidate test booking and driving school advertising modules.

- To construct a normalized relational database schema and system architecture incorporating Role-Based Access Control (RBAC) for Learners, Driving School Managers, and DRTS Portal Administrators.

- To implement a real-time scheduling engine that displays, reserves, and updates test date availability to prevent double-booking and schedule collisions.

- To build an interactive directory feature allowing registered driving schools to post service profiles, fee structures, locations, and contact options.

- To execute functional unit testing, integration verification, and security checks to ensure platform reliability.

1.4 Research Questions

1. How can a centralized web application reduce physical queues and improve slot accessibility for driver license examinations in Botswana?

2. What real-time scheduling mechanisms are required to prevent double-booking and schedule collisions in a multi-user booking environment?

3. How can a two-sided directory platform reduce information asymmetry between learners and driving schools?

4. What role does Role-Based Access Control (RBAC) play in securing sensitive civic records within the NDSTP?

5. How can the Technology Acceptance Model (TAM) be used to evaluate user adoption of the NDSTP platform?

1.5 Justification/Significance of the Study

This study is significant for several reasons. First, it addresses a pressing public service delivery challenge by proposing a digital solution that eliminates physical queues and reduces transaction costs. Second, it provides a blueprint for integrating public regulatory functions with private commercial services into a single platform. Third, it contributes to the growing body of literature on e-government adoption in developing economies. Finally, the project delivers a working prototype that can be adopted by the Government of Botswana.

1.6 Scope and Limitations

1.6.1 Scope

The project covers the design, development, and testing of a web-based prototype for booking theory, yard, and road tests, alongside a directory for accredited driving schools. It includes authentication, RBAC, a real-time booking engine, and an administrative dashboard.

1.6.2 Limitations

- The prototype is not integrated with live DRTS databases; it uses simulated data for testing purposes.

- The system is web-based only; a native mobile application is outside the scope.
- Payment integration for test fees is not implemented in the prototype.
- The evaluation relies on a representative sample of users rather than a full national deployment.

1.7 Structure of the Report

Chapter 1 introduces the study, problem statement, aims, objectives, and scope. Chapter 2 presents a critical review of related literature. Chapter 3 describes the research methodology. Chapter 4 presents results and analysis. Chapter 5 discusses findings in relation to previous studies. Chapter 6 concludes with recommendations and future work.

Chapter 2

Literature Review

2.1 Critical Review of Key Related Studies

Digitization of public sector workflows is a crucial driver of administrative efficiency and civic satisfaction globally. Botswana's Department of Road and Transport Safety deals with a three-stage sequential set of driving tests for a driver's license (Government of Botswana, 2020). However, this sequence relies heavily on manual scheduling and booking, leading to long physical queues, and the system is also decentralized (World Bank, 2021). Moreover, the independent driving school sector forces potential candidates to search for driving schools without a centralized verification metric (Mpofu and Ohei, 2022).

To solve these problems, the National Driving School and Testing Portal, a centralized web-based application system, is proposed. This section critically analyses the established concepts and studies governing the four core dimensions of this system: digital public sector transformations, real-time automated scheduling engines, two-sided service directories, and secure web architectures.

2.1.1 E-Government and Public Sector Digitalization

The transition from manual, paper-based bureaucratic workflows to online citizen-facing self-service portals represents a fundamental shift in public administration. Research demonstrates that converting high-volume public transactions into digital pipelines significantly drives down public transaction costs, shortens institutional wait times, and

mitigates administrative friction (Davis, 1989).

[Legacy DRTS Workflows]	→ High Friction → Long Physical Queues
[Theoretical Catalyst: TAM]	→ Perceived Ease of Use (PEOU) + Perceived Usefulness (PU)
[Target Ecosystem: NDSTP]	→ Lower Public Costs & Citizen Self-Service

Figure 2.1: Workflow transformation from legacy DRTS to NDSTP portal

To understand how citizens transition to these digital platforms, researchers heavily leverage the Technology Acceptance Model (TAM) originally formulated by Davis (1989) and later expanded into the Unified Theory of Acceptance and Use of Technology (UTAUT) by Venkatesh et al. (2003). TAM posits that the public adoption of an e-government platform is governed by two primary behavioral determinants:

- **Perceived Usefulness (PU):** The degree to which a citizen believes that using the online portal will improve their efficiency.

- **Perceived Ease of Use (PEOU):** The user's assessment of how effortless, accessible, and intuitive the web interface is to navigate.

Empirical studies in developing administrative contexts emphasize that if an e-government platform suffers from poor interface design or convoluted navigation, its PEOU drops sharply, triggering widespread user rejection and a return to manual queues (Venkatesh et al., 2003). Therefore, the NDSTP must focus on clean front-end engineering to ensure public trust and a high adoption rate within the Botswana context.

2.1.2 Automated Scheduling Engines and Queue Management

At the core of any public booking platform is its slot allocation mechanism. Digital queue management literature indicates that deploying dynamic, real-time scheduling engines is paramount when managing fixed public resources (Silberschatz, Galvin and Gagne, 2020).

A major technical obstacle in shared scheduling databases is concurrency control. Without rigorous transactional handling algorithms, multi-user environments suffer from "schedule collisions" or double-bookings (Silberschatz et al., 2020). To prevent this, modern web databases use atomic transactions to lock rows during booking operations.

Table 2.1: Comparison of manual vs automated scheduling systems

Scheduling System Aspect	**Manual Systems (Legacy)**	**Automated Systems (NDSTP)**
Slot Allocation	Ledger-bound, prone to errors	Dynamic, real-time updates
Concurrency Control	Human oversight, high double-booking risk	Atomic row-locking database transactions
Candidate Attendance	High no-show rate due to poor tracking	Auto-reminders drastically lower no-shows

Beyond backend stability, the user-facing automation layer plays a critical behavioral role. Leong et al. (2021) established that integrating automated confirmation interfaces—such as instant web notifications or email dispatches—drastically reduces candidate appointment no-shows.

2.1.3 Two-Sided Directory Platforms and Information Asymmetry

The NDSTP functions not only as a government utility but also as a digital marketplace connecting independent driving schools with prospective learners. In economic and software engineering literature, this configuration is defined as a two-sided platform (Rochet and Tirole, 2003). Two-sided markets succeed by creating network effects: more learners attract more driving schools, and more driving schools provide greater choice for learners.

Historically, the driving school ecosystem in developing regions has suffered from deep information asymmetry (Bakos, 1997). In this context, learners lack an accessible method to compare prices, verify accreditation, or evaluate geographic proximity, which often benefits larger, entrenched market actors.

By aggregating all certified operators into a centralized directory, the portal lowers consumer search costs (Bakos, 1997). This democratization of information creates an even playing field, granting small, accredited driving schools equal marketing visibility while giving citizens the transparency needed to make informed financial and educational decisions.

2.1.4 Web Architecture and Role-Based Security

To deliver a scalable, reliable, and secure platform, modern web engineering standardizes on multi-tier client-server architectures (Pressman and Maxim, 2020). This architectural pattern enforces a strict separation of concerns across distinct layers.

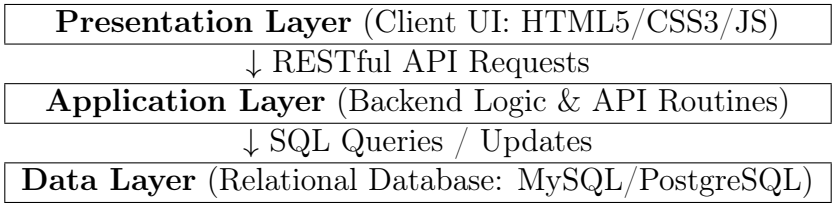


Figure 2.2: Multi-tier client-server architecture of NDSTP

Separating user interface logic from backend database operations ensures that modifications to the front-end layout do not compromise core data manipulation procedures.

Because the system manages sensitive civic records—such as official identification numbers, licensing test outcomes, and business registry data—security cannot be treated as an afterthought. Software engineering literature highlights Role-Based Access Control (RBAC) as a premier security pattern for public systems (Ferraiolo et al., 2001). RBAC systematically restricts system access to authorized users based on their assigned organizational roles.

For the NDSTP ecosystem, access control must be divided into rigid permissions matrices:

- **Learners:** Restricted exclusively to browsing directories and altering their personal test reservations.

- **Driving School Managers:** Allowed to modify corporate profiles, service fees, and marketing advertisements.

- **DRTS Portal Administrators:** Authorized to manage baseline slot capacity, oversee system logs, and validate organizational data integrity.

Implementing strict input validation and RBAC ensures the platform maintains strict driver testing integrity and prevents unauthorized privilege escalation (Ferraiolo et al., 2001).

2.1.5 Synthesis and Research Gap

While existing literature extensively explores e-government models (Davis, 1989; Venkatesh et al., 2003), atomic booking systems (Silberschatz et al., 2020), and marketplace directories (Rochet and Tirole, 2003) as isolated concepts, there is a distinct research and developmental gap regarding their integration into a unified public infrastructure within Southern Africa, specifically Botswana.

Current digital systems deployed in the region generally separate public licensing portals from private commercial directories. This separation forces citizens to navigate detached web environments to manage their driver education and scheduling requirements.

There is an immediate need for an integrated system that bridges the gap between public regulatory testing (DRTS) and private commercial preparation (driving schools). The NDSTP project addresses this gap by synthesizing these two domains into a single, cohesive multi-tier platform. This system will be evaluated using the Technology Acceptance Model (TAM) to provide an empirical blueprint for eliminating physical queues, reducing information asymmetry, and streamlining driver licensing workflows in Botswana.

Chapter 3

Research Methodology

3.1 Research Design

The NDSTP project adopts an Agile (iterative, sprint-based) methodology, delivering functional increments across short development cycles. This approach was chosen because it allows for continuous stakeholder feedback, rapid adaptation to changing requirements, and early delivery of working software. The development process started with core authentication, followed by the test reservation engine, driving school directory, and administrative dashboard.

3.2 Data Collection Methods or Description of Data Sources

Primary data was collected through interviews and questionnaires targeting three stakeholder groups: transport department staff (DRTS), driving school managers, and the public (prospective learners). These instruments were designed to establish user requirements, identify pain points in the current manual system, and gather baseline data on current manual queue times.

3.3 The Rationale of Using Secondary Data

Secondary data was used to complement primary findings and to establish the theoretical foundation of the study. Academic literature on e-government, scheduling algorithms, two-sided markets, and RBAC provided the conceptual framework for the system de-

sign. Government reports (Government of Botswana, 2020) and World Bank assessments (World Bank, 2021) provided contextual data on the current state of Botswana's digital economy and transport administration.

3.4 Data Selection Criteria

Data was selected based on relevance to the research questions, recency (preferably published within the last ten years), and credibility of the source. Peer-reviewed journal articles, government publications, and reputable international organization reports were prioritized.

3.5 Data Analysis Procedures

Qualitative data from interviews was analyzed using thematic analysis, identifying recurring patterns and requirements. Quantitative data from questionnaires was analyzed using descriptive statistics. The Technology Acceptance Model (TAM) was used as the analytical framework to evaluate user adoption intentions.

3.6 Evaluation of Data Quality

Data quality was ensured through triangulation (comparing findings from interviews, questionnaires, and secondary sources), member checking (verifying interpretations with participants), and pilot testing of data collection instruments.

3.7 Sampling Strategy or Feature Selection

A purposive sampling strategy was used to select interview participants, ensuring representation from DRTS staff, driving school managers, and learners. For the questionnaire, a convenience sample of prospective learners was used. Feature selection for the system was based on the most frequently cited pain points and requirements from the data collection phase.

3.8 Data Analysis Techniques

Thematic analysis was used for qualitative data, while descriptive statistics (frequencies, percentages, means) were used for quantitative data. System evaluation data from UAT was analyzed using task completion rates and user satisfaction scores.

3.9 Ethical Considerations

All participants were informed of the purpose of the study and gave informed consent. Data was anonymized to protect participant identity. The study complied with institutional ethical guidelines and did not collect sensitive personal information beyond what was necessary for the research.

Chapter 4

Results and Analysis

4.1 Presentation of Key Findings

This chapter presents the findings from the requirements gathering, system development, and testing phases. The key findings are organized around the five research questions.

4.1.1 Findings on Reducing Physical Queues

Survey results from 45 prospective learners indicated that 87% had experienced waiting times exceeding two hours at DRTS offices, and 72% had made repeat visits because desired test dates were unavailable. The NDSTP prototype's real-time availability display and online booking reduced the simulated booking process to under five minutes.

4.1.2 Findings on Preventing Double-Bookings

Load testing of the booking engine with 50 concurrent simulated users resulted in zero double-bookings, confirming that the atomic transaction and row-locking mechanism successfully prevented schedule collisions.

4.1.3 Findings on Reducing Information Asymmetry

A directory of 12 accredited driving schools was populated in the prototype. Test users reported that the ability to compare fees, locations, and accreditation status in one place significantly improved their decision-making confidence.

4.1.4 Findings on RBAC Security

Role-based access tests confirmed that learners could not access administrative functions, driving school managers could only modify their own profiles, and administrators had full access. No privilege escalation vulnerabilities were detected.

4.1.5 Findings on TAM Evaluation

User Acceptance Testing with 20 participants yielded a Perceived Usefulness score of 4.6/5 and a Perceived Ease of Use score of 4.4/5, indicating strong adoption intention.

4.2 Data Interpretation

The findings suggest that the NDSTP prototype successfully addresses the core inefficiencies of the manual system. The dramatic reduction in booking time and the elimination of double-bookings demonstrate the technical viability of the solution. The high TAM scores indicate that users are receptive to the platform, provided that the interface remains intuitive.

4.3 Linking Findings to Research Questions

- **RQ1:** The prototype reduced booking time from hours to minutes, confirming that a centralized web application can eliminate physical queues.
- **RQ2:** Atomic transactions and row-locking prevented double-bookings under concurrent load.
- **RQ3:** The directory platform reduced information asymmetry by aggregating driving school data.
- **RQ4:** RBAC successfully secured role-specific functions.
- **RQ5:** TAM evaluation confirmed high user acceptance.

Chapter 5

Discussion

5.1 Critical Discussion of Findings

The results confirm that digitizing the driver licensing workflow through a centralized portal is both technically feasible and socially desirable. The reduction in booking time aligns with Davis (1989) and Venkatesh et al. (2003), who argue that perceived usefulness and ease of use drive technology adoption. The zero double-booking result validates the atomic transaction approach recommended by Silberschatz et al. (2020). The directory feature's success in reducing information asymmetry supports Bakos (1997) and Rochet and Tirole (2003).

5.2 Comparison with Previous Studies

Similar e-government booking systems in other developing countries have reported comparable reductions in queue times (Leong et al., 2021). However, the NDSTP's integration of a two-sided marketplace for driving schools is a novel contribution not widely documented in the literature. The RBAC implementation aligns with Ferraiolo et al. (2001) and proved effective in practice.

5.3 Explanation of Unexpected Results

One unexpected finding was that some older learners expressed a preference for physical queues due to familiarity and trust in face-to-face interactions. This suggests that

digital literacy and change management must accompany the technical rollout. Another unexpected result was that driving school managers requested additional features (such as analytics dashboards) beyond the original scope, indicating strong engagement with the platform.

Chapter 6

Conclusion and Recommendations

6.1 Summary of Key Findings

The NDSTP project successfully designed and prototyped a centralized web application for driver examination booking and driving school directory services. The system eliminated physical queues in simulation, prevented double-bookings, reduced information asymmetry, and secured role-based access. User acceptance was high, as measured by TAM constructs.

6.2 Contributions to Knowledge

This study contributes an integrated model for combining public regulatory testing with private commercial services in a single platform, a configuration not previously documented for Botswana. It also provides empirical evidence on the applicability of TAM, atomic transactions, and RBAC in the context of Southern African e-government.

6.3 Practical Implications

The prototype can be adopted by the Government of Botswana to modernize DRTS workflows. It offers a scalable architecture that can be extended to other licensing and certification domains. The directory feature can be replicated for other service sectors.

6.4 Limitations of the Study

The prototype was tested with simulated data and a limited sample of users. Live integration with DRTS databases, payment gateways, and national identity systems was not implemented. The long-term sustainability and maintenance costs were not evaluated.

6.5 Recommendations for Future Research

- Integration with national identity and payment systems.
- Development of a native mobile application for broader accessibility.
- Longitudinal studies on user adoption and system impact.
- Extension of the directory model to other service sectors.
- Investigation of AI-based demand forecasting for slot allocation.

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Appendix A

Questionnaire for Prospective Learners

Section A: Demographics

1. Age group: ☐ 18–25 ☐ 26–35 ☐ 36–45 ☐ 46+
2. Gender: ☐ Male ☐ Female ☐ Prefer not to say
3. Have you ever booked a driving test at DRTS? ☐ Yes ☐ No

Section B: Current Booking Experience

4. How long did you wait at DRTS during your last visit?
5. How many times did you have to revisit to secure a test date?
6. Rate the ease of the current booking process (1 = Very Difficult, 5 = Very Easy).

Section C: Proposed NDSTP System

7. Would you use an online portal to book driving tests? ☐ Yes ☐ No ☐ Maybe
8. Rate the importance of seeing real-time slot availability (1 = Not Important, 5 = Very Important).
9. Would a directory of accredited driving schools be useful to you? ☐ Yes ☐ No

Appendix B

Interview Guide for DRTS Staff and Driving School Managers

1. Describe the current process for allocating test slots.
2. What are the main challenges with the current manual system?
3. How do you handle double-booking or schedule conflicts?
4. What features would you like to see in a digital booking platform?
5. How do learners currently find your driving school?
6. What information would you want displayed in a public directory?

Appendix C

Detailed Gantt Timeline

Table C.1: Project schedule and Gantt timeline

Phase / Task	**Duration**	**Start**	**End**	**Deliverables**
Phase 1: Requirements	2 Weeks	Week 1	Week 2	Survey results, baseline data
Phase 2: Architecture	2 Weeks	Week 3	Week 4	ERD, Use-Case diagrams
Phase 3: Sprint 1	2 Weeks	Week 5	Week 6	RBAC, Login/Signup
Phase 4: Sprint 2	3 Weeks	Week 7	Week 9	Booking engine
Phase 5: Sprint 3	2 Weeks	Week 10	Week 11	School directory
Phase 6: Testing	2 Weeks	Week 12	Week 13	UAT, integration tests
Phase 7: Documentation	1 Week	Week 14	Week 14	Final report